Digital Twin Architecture Pipeline

End-to-end data processing and dashboard visualization workflow

Simulated Patient Data

STEP 01

Generates real-time synthetic physiological parameters and continuous health data streams.

↓

Python Processing

STEP 02

Utilizes the **Pandas Framework** for data cleaning, transformation, aggregation, and time-series normalization.

↓

Digital Twin State

STEP 03

Maintains updated state models representing current real-time patient health metrics and clinical status.

↓

Streamlit Dashboard

STEP 04

Interactive web platform presenting real-time patient analytics and key indicators:

- Sleep Duration
- Physical Activity
- Heart Rate
- Blood Glucose

System Overview

This flow highlights a scalable pipeline that ingest simulated patient inputs, processes telemetry through Python/Pandas logic, computes digital twin state vectors, and renders actionable medical metrics via an interactive Streamlit UI.